ADITYA MAJUMDAR

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Ph.D. Student in Informatics ◇ Penn State University ◇ State College, PA, USA

RESEARCH INTERESTS

AI for Social Impact, Human-AI Interaction, AI Evaluation and Measurement, Responsible AI, Human-Centered AI, Algorithmic Fairness

EDUCATION

Pennsylvania State University (*GPA*: 4.0/4.0) **State College, US**
Ph.D. in Informatics *Aug 2024 - Present*
Advisor: Prof. Amulya Yadav

Pennsylvania State University (*GPA*: 3.86/4.0) **State College, US**
B.S. in Computer Science (Minor in Engineering Leadership Development) *Aug 2020 - May 2024*
Cum Laude

PEER-REVIEWED PUBLICATIONS

Conference Papers

[1] **Aditya Majumdar**, Wenbo Zhang, Kashvi Prawal, Amulya Yadav. *The Hardness of Achieving Impact in AI for Social Impact Research: A Ground-Level View of Challenges & Opportunities.* In Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2026.

[2] Bonam Mingole, **Aditya Majumdar**, Firdaus Choudhury, Amulya Yadav. *Dr. GPT Will See You Now, but Should It? Exploring the Benefits and Harms of Large Language Models in Medical Diagnosis using Crowdsourced Clinical Cases.* In Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2026.

Under Review

[3] **Aditya Majumdar**, Arsh Gupta, Devanshi Gupta, Hongjin Lin, Ayan Mukhopadhyay, Amulya Yadav. *SPARK: Co-creating a Reflection Toolkit for Early Stage AI for Social Impact Collaborations.* Under review at FAccT 2027.

[4] Wenbo Zhang, **Aditya Majumdar**, Amulya Yadav. *CHAI for LLMs: Improving Code-Mixed Translation in Large Language Models through Reinforcement Learning with AI Feedback.* Under review at FAccT 2027.

[5] Qian Ma, **Aditya Majumdar**, Sarah Rajtmajer, Brett Frischmann. *Consent without Comprehension: A Randomized Experiment on Pedagogical Friction in Privacy Policy Flows.* Under review at CHI 2027.

Workshop Papers

[6] Qian Ma, Yingfan Zhou, Shubhang Kaushik, Aamod Joshi, **Aditya Majumdar**, Noah Apthorpe, Yan Shvartzshnaider, Sarah Rajtmajer, Brett Frischmann. *Pedagogical Friction in Password Creation.* In IEEE ACSAC Workshop on Cyber Security Experimentation and Test (CSET), 2026.

Working Papers

[7] Firdaus Choudhury, **Aditya Majumdar**, Arunesh Sinha, Amulya Yadav. *From Manual Planning to Automated Routing: Scalable Sapling Distribution for Smallholder Farmers in India.*

[8] Bonam Mingole, Samarth Khanna, **Aditya Majumdar**, Janice Ahn, Humaira Farid, Amandeep Singh, Suman Saha, Amulya Yadav. *Effectiveness of LLMs in Education.*

[9] **Aditya Majumdar**, Maria Vinca, Sarah Rajtmajer. *Moral Sycophancy in AI Chatbots and User Wellbeing.*

MEDIA COVERAGE

- "Dr. Google" had its issues. Can ChatGPT Health do better? **MIT Technology Review** - Article *Feb 2026*

SELECTED AWARDS & FELLOWSHIPS (BY YEAR)

- **LinDiv Fellowship** | NSF NRT, Penn State University *2025*
  Selected for an NSF-funded traineeship advancing human-centered technology for linguistically diverse populations through interdisciplinary research and training.
- **DAAD RISE Award** | German Academic Exchange Service *2023*
  Selected as one of 280 students from 1,700 applicants to conduct research in Germany.
- **Fotis Scholarship** | Penn State University *2023*
  Awarded to 5 Residential Assistants out of 300 for outstanding leadership and scholastic achievement.
- **Mike Erdmann Global Leadership Award** | SEDTAPP, Penn State University *2023*
  Sole recipient among 35 students in the ELD minor, for commitment to professional excellence.

RESEARCH & INDUSTRY EXPERIENCE

Pennsylvania State University, Research Assistant, PA, US

Research Topic: Moral Sycophancy in AI Chatbots and User Wellbeing *May 2026 - Present*
Advisor: Prof. Sarah Rajtmajer

- Designing a human-subjects study of how chatbot validation of one-sided accounts of interpersonal conflicts affects user wellbeing, combining a survey with participant-submitted chat logs.
- Built a Qualtrics survey with a chat-log upload workflow and a battery of validated measures covering loneliness, social support, empathy, and problematic internet use.
- Developed a Chrome extension that lets participants export and redact their chat logs locally before submission.
- Developing an LLM-based sycophancy detection pipeline adapted from published codebooks and benchmarks, with a power analysis and a multilevel regression plan for the study.

Research Topic: SPARK, a Reflection Toolkit for Early-Stage AI-Nonprofit Collaborations *Nov 2025 - Present*
Advisor: Prof. Amulya Yadav

- Ran four focus groups with 15 AI researchers and partner-organization staff to identify what each side needs to know before deciding to collaborate.
- Designed SPARK, a pair of structured reflection forms (47 questions) that surface each side's needs and expectations before a project begins, and refined them using participant feedback (12 of 15 responded; average usefulness 4.5 out of 5).
- Evaluated SPARK against a current baseline in a randomized study ($n = 71$): in the writing task ($n = 36$), SPARK scored higher on clarity and usability and 78% preferred it; in the reading task ($n = 35$), ratings were similar and 63% preferred it.

Research Topic: Geospatial Optimization for Restorative Agriculture Services *Aug 2025 - Present*
Advisor: Prof. Amulya Yadav *Partner*: Naandi Foundation, India

- Partnering with the Naandi Foundation, a nonprofit supporting farmers moving to restorative agriculture, to plan sapling delivery across rural regions.
- Formulating mixed-integer programming models for a vehicle routing problem under real-world logistical constraints.
- Analyzing geospatial field data to reduce travel time and operating cost while increasing service coverage.
- Working with Naandi staff so that model outputs match field-level realities.

Research Topic: Code-mixing Abilities of LLMs *Sep 2023 - Present*
Advisor: Prof. Amulya Yadav

- Developed a reinforcement learning with AI feedback approach to improve code-mixed translation in LLMs.
- Fine-tuned LLMs, including GPT-4o, on code-mixed data using the Hugging Face framework.
- Wrote Python pipelines to preprocess training and test data.
- Deployed and tuned models on Google Cloud Platform.

Research Topic: Barriers to Deployment in AI for Social Good *Jan 2025 - Jan 2026*
Advisor: Prof. Amulya Yadav

- Conducted 26 semi-structured interviews with AI researchers and practitioners to examine why many AI for Social Good projects fail to deploy.

- Applied qualitative coding and thematic analysis to identify misaligned academic incentives, stakeholder collaboration challenges, and gaps between prototyping and field deployment.
- Findings published at FAccT 2026, with recommendations for more context-sensitive design and evaluation practices.

Research Topic: Experimental Design for Online Safety and Consent Mechanisms *May 2025 - Sept 2025*
Advisor: Prof. Sarah Rajtmajer

- Designed and implemented an experimental consent mechanism to improve informed parental consent on children's online platforms.
- Developed randomized experimental protocols using A/B testing to compare alternative consent and privacy policy presentations.
- Built backend infrastructure to securely store participant responses and behavioral interaction data.
- Worked with interdisciplinary researchers to turn experimental design decisions into deployable web implementations.

Research Topic: Equitable Resource Allocation for Homeless Populations *Mar 2025 - Jan 2026*
Advisor: Prof. Amulya Yadav *Partner*: Allegheny County Department of Human Services, USA

- Worked with Allegheny County Department of Human Services (ACDHS) to estimate wait times in housing assignment for people experiencing homelessness, to improve transparency and access.
- Ran exploratory data analysis and built models on demographic, housing, and queue data to predict assignment delays.
- Evaluated fairness criteria (e.g., demographic parity) to find and reduce disparities in predicted wait times across marginalized subgroups.
- Worked with caseworkers to interpret findings and fit model outputs into operational workflows.

Research Topic: AI for Public Health *Jan 2025 - May 2025*
Advisor: Prof. Amulya Yadav

- Preprocessed physician-labeled medical datasets to evaluate AI-generated responses with LLMs (e.g., GPT-4).
- Designed few-shot prompts and varied in-context example sizes to test model sensitivity and consistency.
- Built automated LLM pipelines that replicate physician grading of AI responses, using a majority-voting ensemble and ROC-based threshold selection.
- Measured the reliability of LLM evaluations with F1, precision, recall, and AUC.

Research Topic: Technology-Enabled Case Management for Child Protection *Aug 2024 - Mar 2025*
Advisor: Prof. Amulya Yadav *Partner*: World Hope International, Philippines

- Collaborated with World Hope, a nonprofit supporting children affected by online exploitation in the Philippines, a region identified as a global hotspot for online child sexual abuse.
- Led weekly meetings with non-technical NGO staff, explaining technical concepts in plain terms, and gathered requirements to match the NGO's objectives at minimal deployment cost.
- Built a Flutter application with an AppSheet database that serves as a low-cost digital intake and case management system, reducing documentation burden for frontline staff.

DAAD RISE Scholar | University of Konstanz, Germany *May 2023 - Aug 2023*
Research Topic: Cyber-Physical Systems
Supervisor: Eduard Buss

- Wrote Matplotlib, Pandas, and NumPy scripts for time series analysis, with a lowpass Butterworth filter for signal cleaning.
- Designed 3D models for multiple sensors and built three experimental assemblies.
- Built a synchronized multi-device system using a hub-and-spoke architecture with Lsyncd and Rsync.

Software Engineering Intern | LegalFundi, Cape Town, South Africa *May 2022 - Jul 2022*
Supervisor: Daniel Nel

- Built the React.js and CSS homepage and chatbot interface for the company website.
- Presented weekly progress updates to managers using OKRs, Gantt charts, and PowerPoint.

TEACHING EXPERIENCE

Pennsylvania State University

- **Teaching Assistant** | DS/CMPSC 442: Introduction to Artificial Intelligence *Jan 2026 - May 2026*

- **Learning Assistant** | ENGR 408: Engineering Leadership Principles *Jan 2022 - May 2024*
- **Learning Assistant** | ECON 104: Introduction to Macroeconomics *Jan 2021 - May 2021*

PROFESSIONAL SERVICE

- **Center for Socially Responsible AI Student Leader** *Feb 2025 - Present*
  Organizer of the CSRAI Young Achievers Symposium and AI for Social Impact Series

TECHNICAL SKILLS

Programming Languages	Python, C, Java, JavaScript, HTML, CSS, SQL
Frameworks & Tools	NumPy, PyTorch, Matplotlib, Pandas, Scikit-learn, Hugging Face, React, Bootstrap, Flutter, AppSheet, Google Cloud Platform, Qualtrics, Figma
Research Methods	Semi-structured interviews, thematic coding, survey and psychometric design, randomized experiments and A/B testing, power analysis, participatory design, LLM-as-judge, mixed-integer programming